

BEATS: Audio Pre-Training with Acoustic Tokenizers

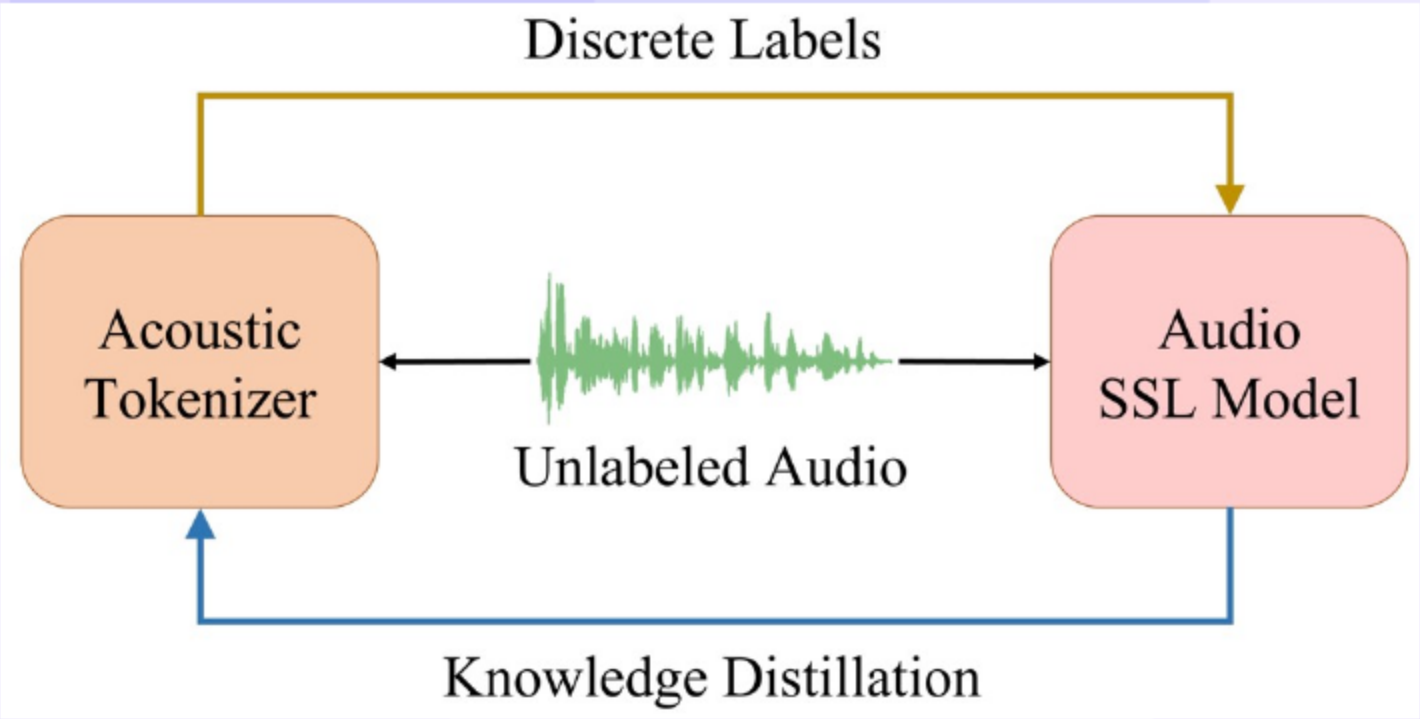
Sanyuan Chen, Yu Wu, Chengyi Wang, Shujie Liu, Daniel Tompkins, Zhuo Chen, Wanxiang Che, Xiangzhan Yu, Furu Wei

01 BEATS: Audio Pre-Training with Acoustic Tokenizers

- Authors
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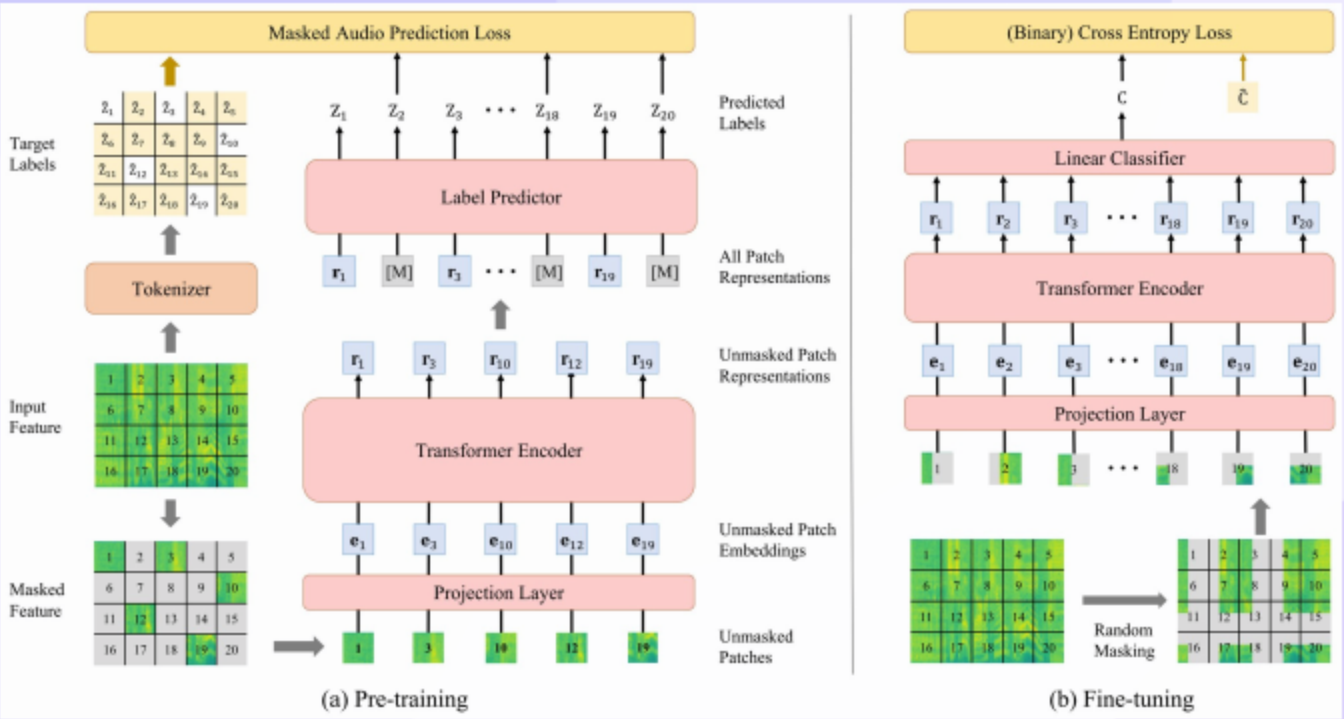
Motivation and Thesis: Discrete-Label Audio SSL with Iterative Tokenizers



- We emphasize semantic abstraction via tokens
- We tackle diverse-audio tokenization
- We improve token quality iteratively
- Discrete labels transfer across domains
- We inject semantics via teacher distillation.
- We unify with discrete labels across modalities.

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Positioning: From Reconstruction/Contrastive SSL to Masked Label Prediction



- We move beyond contrastive/reconstruction
- We predict masked discrete labels
- We distill tokenizers from our model

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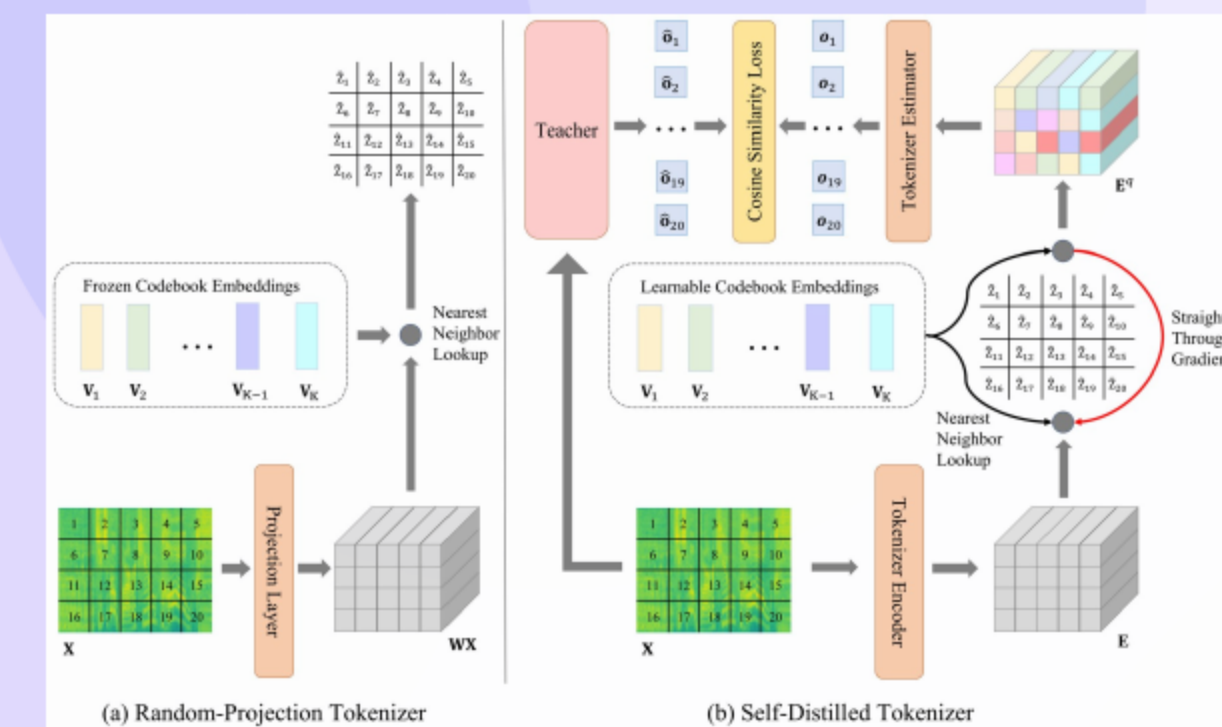
Baselines and Citations: Prior SSL and Tokenizer-Based Methods

- wav2vec 2.0 — Baevski et al., NeurIPS 2020
- HuBERT — Hsu et al., ICLR 2021
- MAE-style masked reconstruction (AudioMAE/MAE) — He et al., CVPR 2022; Huang et al., NeurIPS 2022
- PaSST — Koutini et al., ICASSP 2022
- AST — Gong et al., ICASSP 2021
- HTS-AT — Chen et al., ICASSP 2022
- PSLA — Gong et al., Interspeech 2021
- We report single-model results; see papers for ensembles.

Baevski et al., 2020; Hsu et al., 2021; Koutini et al.

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Iterative Loop and Tokenizers: From Cold Start to Self-Distillation



- We progressively align tokens with semantics
- We alternate labeling and masked prediction

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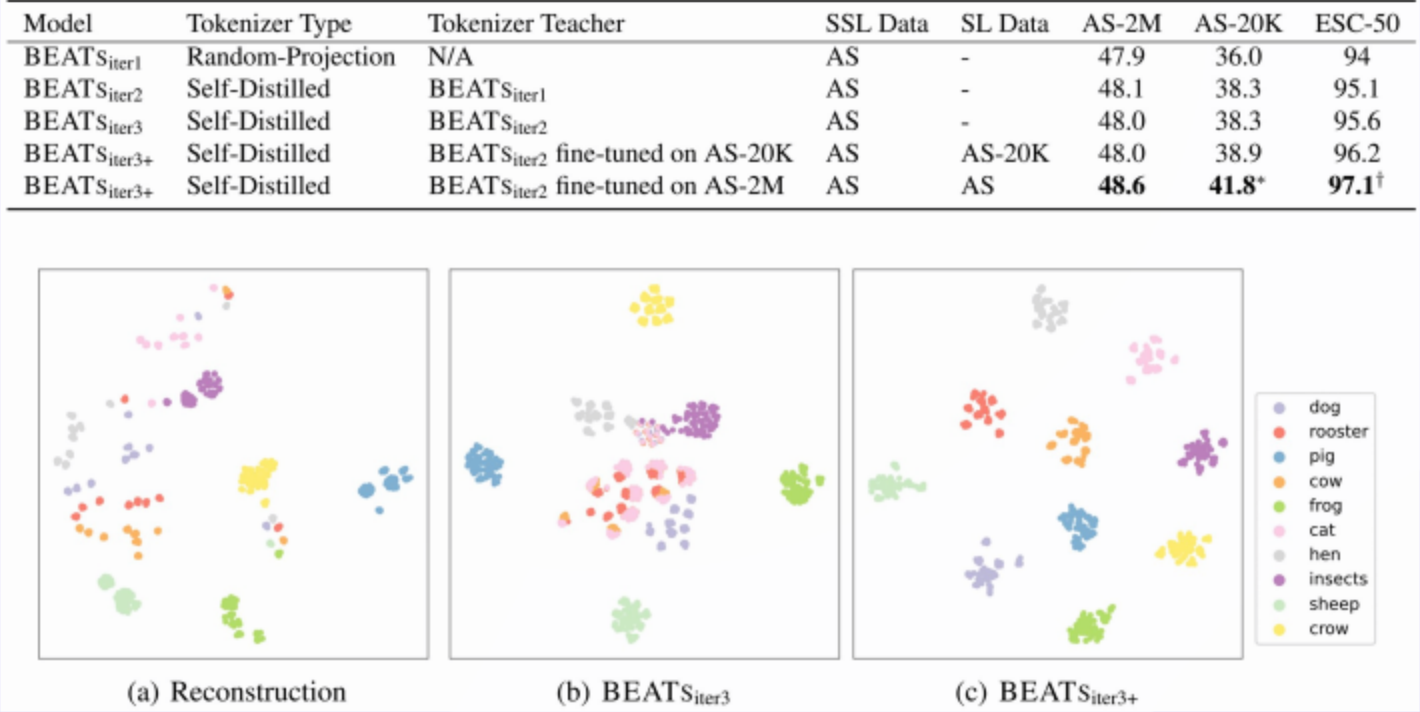
SSL Model and Minimal Setup: ViT + **75%** Masking; Data/Model/Iterations

- We use a ViT backbone with efficient relative bias
- We predict 75% masked labels via cross-entropy

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Results: SOTA and Iteration Gains (incl. teacher effects)

- We achieve 48.6 mAP on AS-2M
- We improve 47.9 → 48.1 → 48.6
- We match larger models: 90M vs ~300M (Koutini '21; Gong '21)



- Note: ignore the small top table in the image; we focus on the t-SNE comparison
- We compare reconstruction vs BEATS (self-supervised teacher) vs BEATS+ (supervised teacher) t-SNEs.

- We deliver state-of-the-art audio SSL
- We note compute and data limits
- We plan larger, multimodal scaling

Thank You

Questions & Discussion